

Expect legal challenge if Eskom sticks with its irrational procurement course

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Eskom is becoming a problem. It sells less electricity than it did 10 years ago, yet it employs 50% more staff, its coal costs have trebled, and its tariffs and revenues have increased fourfold. Now Eskom is blocking cheaper independent power producers (IPPs) and is attempting to mislead the government and the public by arguing that renewable energy and gas power are too expensive, and that Eskom should fast-track the procurement of nuclear energy.

Eskom was a problem before, under apartheid. It served mainly white South Africans. And in the previous century, it was also not particularly efficient in building new power stations. There were cost and time overruns, and there were periods with unacceptably high increases in electricity tariffs.

However, at the birth of our democracy, Eskom was in a sweet spot. It had surplus generation capacity, which was mostly paid for. Its short-run marginal cost of production was low. It entered into two multiyear price compacts with the government to reduce the real price of electricity. Eskom also helped deliver the most successful component of the governing party's reconstruction and development programme by increasing the proportion of South Africans with access to electricity from just more than a third to nearly 90% today.

But between 2006 and 2015, power cuts constrained economic growth, employment and poverty reduction. Now, Eskom is no longer affordable and it is frustrating, rather than supporting, the development of the economy. It is proving to be a poor instrument for a developmental state.

The big questions and decisions for the future are around investment in new power. The electricity business is very capital-intensive, more so than any other infrastructure sector. Eskom's balance sheet, investments and revenues dwarf those in Transnet, Telkom or any other state-owned enterprise. It's crucial that investment decisions are least-cost and that capital is executed efficiently.

It is 12 years since then public enterprises minister Alec Erwin instructed Eskom to build new coal power stations. Medupi and Kusile will be

completed only in the early 2020s, nearly 20 years later. Costs have more than doubled and will likely end up being three times over budget.

During this time, Eskom lost its treasured investment-grade rating and its bonds are now rated junk. As a result, Eskom's capital raising is a lot more challenging and expensive. Eskom has 10 times more debt than a decade ago, and its new coal power stations are not even half-built.

Within this context, it is flabbergasting that Eskom now plans to rush the procurement of a fleet of new nuclear power stations. It claims we need large, additional increments of power soon, that nuclear energy is the cheapest option and that it can easily be financed. All of these claims are wrong.

SA's electricity plan — the Integrated Resource Plan (IRP) — is being updated by the Department of Energy. It incorporates new projections on electricity demand and new comparative costs for supply options. As reported in this newspaper, the new plan suggests that nuclear power might only be required in the 2030s — in other words there is no rush to start a procurement process now.

Crucially, the computer model informing the new electricity plan picks nuclear energy only if the contribution of renewable energy is constrained — in other words if a cap is placed on how much renewable energy can be built in any year. Solar and wind energy are now so cheap that the model picks these supply options first, along with gas power to ensure system reliability.

SA has favourable solar and wind resources, and land is not a constraint. Other countries have built much more renewable energy than we plan to. The constraints in the updated plan regarding the contribution of solar and wind, combined with gas, are thus entirely artificial.

The modelling tools used by staff supporting the department are cutting edge and widely used internationally. Research institutions in SA have the same software. They can mimic the demand forecasts and cost assumptions made by the department and can model similar scenarios and energy mixes; indeed they are already doing so. It will not be very clever for the department or Eskom to now amend or subvert the new IRP.

It is crystal-clear: a least-cost electricity plan for SA does not include nuclear energy, even if a carbon emissions cap is imposed that would enable us to meet our climate change mitigation obligations. Thus if Eskom proceeds with the procurement of nuclear power stations, it will not be a rational decision

and almost certainly will be subject to legal challenge. The original decision by the Cabinet to opt for nuclear energy is already in the courts.

Eskom executives have launched a concerted public relations campaign in recent months, arguing that renewable energy is too expensive. They are correct that the first generation of IPPs in SA were costly, but they omit the inconvenient fact that prices for solar and wind in the latest IPP bid rounds have dropped dramatically and are now down to 62c/kWh, cheaper than Eskom's average cost of supply at 84c/kWh and far cheaper than Medupi or Kusile or new nuclear power. The recent coal IPPs, announced by the department, are also cheaper than Eskom's new build.

The debates about the relative costs of different power sources can be settled if we look at actual power contracts. Fortunately, the department's IPP programme is transparent: private companies have signed 20-year contracts at fixed prices, adjusted only by inflation. There can thus be no controversy about the costs of solar and wind energy. They are now the cheapest sources of grid power. Perhaps the only question is how cheap. Recent renewable energy auctions in Peru, Mexico, Chile and Dubai have seen prices for wind and solar fall even further.

Of course, we should not consider only the costs of individual supply technologies, but also the system costs to ensure reliable supply. Solar and wind resources are variable and need to be complemented with flexible power sources such as gas, hydroelectricity, storage or demand-side management. The blended cost of solar plus wind plus flexible system resources is cheaper than Eskom coal or nuclear.

Choices about SA's future energy mix are far too important to be made by a couple of Eskom executives. The updated IRP should be published and open to public scrutiny and consultation.

Eskom now needs to be restructured. It is intolerable that it is frustrating the entry of IPPs by delaying and inflating the cost of transmission connections and refusing to sign contracts. Eskom should not be in the position where it is building its own generation capacity and is also the arbiter of whether private investors can enter the market or not. It cannot be a player and a referee.

As I have argued previously on these pages, Eskom's generation business needs to be hived off into a separate state-owned company. Then the residual wires business, along with the system operator, could provide a neutral platform for electricity planning and procurement of least-cost power. This form of power-sector restructuring is the norm in most

industrialised countries and emerging economies, and many have undertaken further reforms that allow private generators to trade in power exchanges or directly with customers.

The Cabinet and Eskom's board would do well to pause before rushing into the procurement of new nuclear power plants. South Africans are watching and they will challenge such a decision. Now is the time to reform our power sector.

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